

Banking Fraud Detection and Risk Analysis Using SQL

Project Overview :

This project analyzes banking transaction data using SQL to identify fraud patterns and risk indicators. The dataset includes transaction details such as transaction amount, payment channel, authentication type, device risk score, anomaly score, card presence, suspicious IP status, and fraud flag. The goal is to explore which factors are associated with higher fraud risk and generate useful insights for fraud monitoring.

Dataset Description :

Item	Description
Database Name :	banking_fraud_db
Table Name :	banking_transactions
Number of Records :	10,000 transactions
Main Target Column:	fraud_flag
Tool Used:	MySQL / MySQL Workbench
Analysis Type:	Fraud Detection and Risk Analysis

Column	Meaning
transaction_id :	Unique ID for each banking transaction
transaction_amount :	Amount of the transaction
payment_channel :	Channel used to perform the transaction
authentication_type :	Type of authentication used
device_risk_score :	Risk level of the device
anomaly_score :	Measures how unusual the transaction is
transaction_velocity_score :	Measures speed or intensity of transaction activity
card_present_flag :	Indicates whether the physical card was present
suspicious_ip_flag :	Indicates whether the IP address was suspicious
fraud_flag :	Indicates whether the transaction is fraudulent

Key Insights :

1. *The dataset contains 10,000 banking transactions.*
2. *Fraudulent transactions represent around 12.51% of all transactions.*
3. *Fraud risk differs across payment channels.*
4. *Authentication type affects fraud percentage.*
5. *Transactions without a physical card may show different fraud behavior compared with card-present transactions.*
6. *Higher anomaly scores are strongly associated with higher fraud risk.*
7. *Suspicious IP addresses are linked to a higher likelihood of fraud.*
8. *Combining anomaly score and device risk score helps identify the riskiest transactions*

Conclusion :

This SQL project demonstrates how transaction data can be analyzed to detect fraud patterns and risk indicators. The analysis showed that fraud is not evenly distributed across all transactions. Some factors, such as anomaly score, suspicious IP status, authentication type, and device risk score, can help identify higher-risk transactions. SQL was used to summarize, compare, group, and rank transactions to support fraud detection decisions.

Files Included :

File	Description
banking_transactions2.csv :	Original dataset
banking_fraud_analysis.sql :	SQL queries used for analysis
Banking_Fraud_Analysis_Report.pdf :	Final project report